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MALAYA**

WQD7001 PRINCIPLES OF DATA SCIENCE

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GROUP ASSIGNMENT

Project Title: Cardiovascular Disease Prediction

GROUP 2 MEMBER'S NAME	MATRIC NO	ROLE
LOO LING YAN	23094683	LEADER
CHU JING HAN	23116920	MAKER
HAU JIA QI	17204762	ORACLE
LOH JING WEI	23118574	DETECTIVE
GOH WEE KEN	23118928	SECRETARY

LECTURER: Assoc. Prof. Dr. Maizatul Akmar binti Ismail

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1.0 Project background

This project focuses on cardiovascular disease prediction using machine learning. The aim is to uncover unseen patterns, derive meaningful insights, and support informed business decisions. The topic of cardiovascular disease prediction is selected due to ischemic heart disease being the second leading cause of death in Malaysia. According to the Department of Statistics Malaysia (DOSM), ischemic heart disease accounted for 18,121 deaths, or 15% of medically certified deaths in 2023.

Organization: Insurance Company

Target Users: New Applicants for Policyholder

Potential Benefits: Assessing the risk profile of policyholders during application to mitigate the cost of medical insurance premiums, setting appropriate premiums if the applicant is low risk, and faster application approval duration.

Data Product: Web-based platform to predict the risk level of cardiovascular disease during the application process.

2.0 Problem Statement

Currently, insurance companies face the challenge of inaccurate risk assessments due to relying on traditional methods. These methods struggle with processing the vast amounts of data stored in data warehouses, leading to time-consuming application approvals. As a result, this inefficiency hampers business growth.

3.0 Project Scope

The scope of this project is to enable insurance companies to effectively predict cardiovascular disease during risk level assessments by utilizing a prediction model. This will streamline the application process and shorten approval duration.

4.0 Objectives

- **To identify important features in causing cardiovascular disease:** Conduct correlation analysis to uncover significant relationships in the dataset, focusing on risk factors associated with cardiovascular disease.
- **To develop machine learning algorithms in detecting cardiovascular disease:** Develop models to detect cardiovascular disease risk level during the initial insurance application process for policyholders.

- **To evaluate performance of different machine learning models:** Assess various machine learning models using diverse performance metrics to determine the best model for detecting cardiovascular disease.
- **To improve cardiovascular disease risk assessment:** Enhance risk level assessments during the application process by utilizing the prediction model and shorten approval duration

5.0 Methodology

5.1 Obtain

The dataset was obtained from Kaggle and consists of factual information about the results of medical examinations provided by patients.

Data source: <https://www.kaggle.com/datasets/sulianova/cardiovascular-disease-dataset>

Feature	Data type	Details
Age	Numeric (int)	Days
Height		cm
Weight	Numeric (float)	kg
Gender	Categorical	1: Female; 2:Male
Systolic blood pressure	Numeric (int)	mmHg
Diastolic blood pressure		mmHg
Cholesterol	Ordinal	1: Normal; 2: Above Normal; 3: Well Above Normal
Glucose		1: Normal; 2: Above Normal; 3: Well Above Normal
Smoking	Binary	0: Non-smoker; 1: smoker
Alcohol intake		0: Does not consume; 1: Consumes
Physical activity		0: Inactive; 1: Active
Cardiovascular disease		0: No CVD; 1: Presence of CVD

5.2 Scrub

To ensure the dataset is clean and ready for analysis, first, remove duplicates based on unique identifiers like patient ID. Next, remove the patient IDs from the dataset. Then, check for any missing values, and if they exist, use median imputation to address them and reduce the impact of missing data.

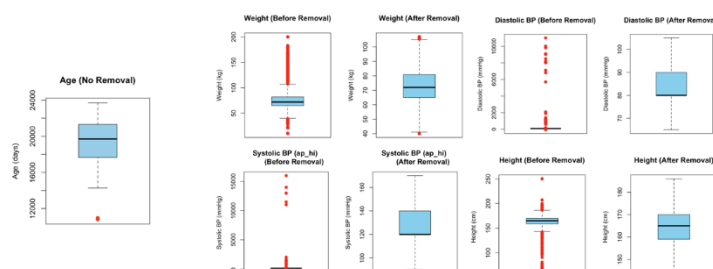
Data transformation is needed to convert numbers of age from days to years.

Before transformation			After transformation		
age	gender	height	age	gender	height
18393	2	168	50	2	168
20228	1	156	55	1	156
18857	1	165	52	1	165
17623	2	169	48	2	169
17474	1	156	48	1	156
21914	1	151	60	1	151

Data normalization on Age, Height, weight, Systolic blood pressure (ap_hi), Diastolic blood pressure (ap_lo).

age	gender	height	weight	ap_hi	ap_lo	cholesterol
-0.4886204	2	0.44344889	-0.8487350	-0.122181107	-0.08823787	1
0.2510059	1	-1.01816077	0.7489360	0.072609644	-0.03517974	3
-0.1927699	1	0.07804648	-0.7098071	0.007679393	-0.14129600	3
-0.7844709	2	0.56524970	0.5405441	0.137539894	0.01787839	1
-0.7844709	1	-1.01816077	NA	-0.187111357	-0.19435413	1
0.9906322	1	-1.62716479	-0.5014152	-0.057250857	-0.08823787	2

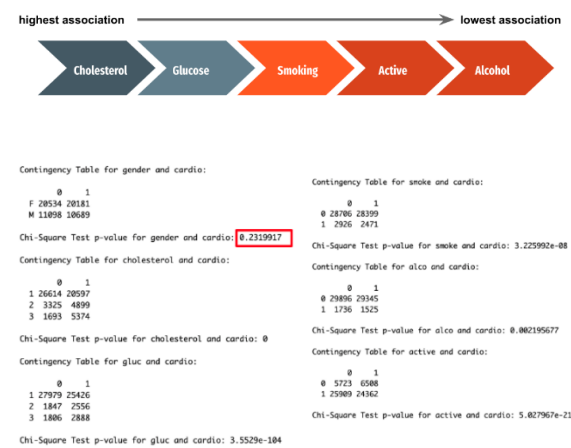
Finally, remove records with outliers by using Interquartile Range (IQR) method.



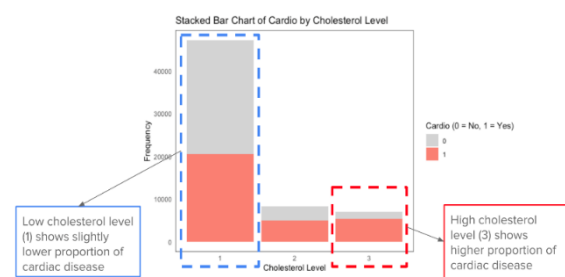
5.3 Exploratory Data Analysis (EDA)

After data preprocessing, exploratory data analysis was conducted.

Qualitative Variables

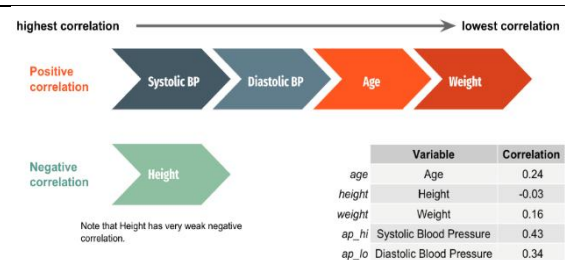


Chi-square test to assess associations between qualitative variables and cardiovascular disease and the computation result. The result shows that the strongest association is Cholesterol, followed by glucose, smoking, activity, and alcohol. In contrast, gender has no significant association with cardiac disease (p-value >0.05).

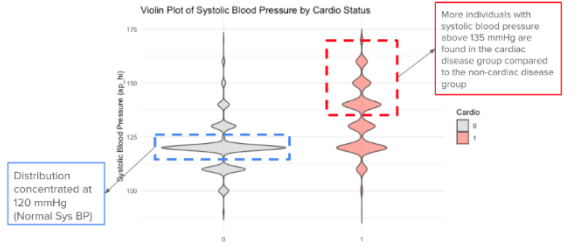
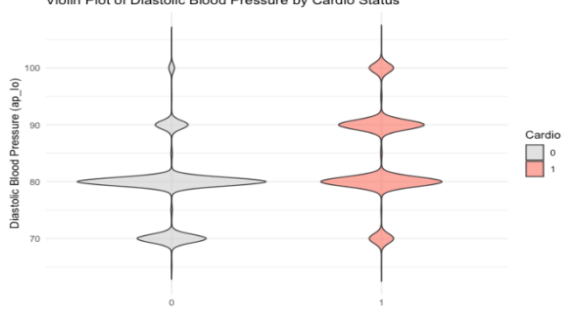
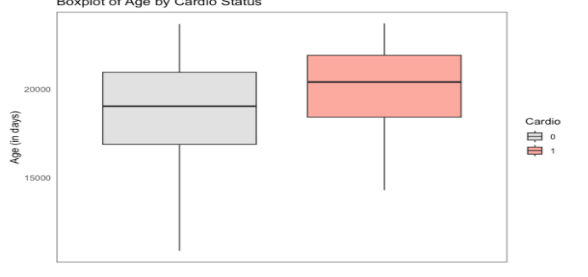
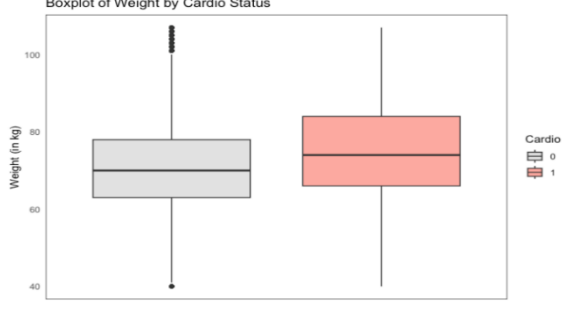


Cholesterol level which has highest association is shown in stacked bar chart, individuals with a low cholesterol level (1), the proportion of those without cardiovascular disease is slightly higher than those with disease. Whereas, for individuals with a high cholesterol level (3), the majority are observed to have cardiovascular disease.

Quantitative Variables



Pearson Correlation to assess relationship between quantitative variables and cardiovascular disease. It shows moderate positive correlations between cardiovascular disease and variables like age, weight, systolic BP, and diastolic BP, while height showed a weak negative correlation.

 Violin Plot of Systolic Blood Pressure by Cardio Status More individuals with systolic blood pressure above 135 mmHg are found in the cardiac disease group compared to the non-cardiac disease group Distribution concentrated at 120 mmHg (Normal Sys BP) Systolic Blood Pressure (mmHg) Cardio 0 1	Violin plot shows systolic blood pressure by cardiovascular disease status. Those without cardiovascular disease are concentrated around 120–130 mmHg, while those with cardiovascular disease show greater variability, with more cases with high blood pressure above 135 mmHg.
 Violin Plot of Diastolic Blood Pressure by Cardio Status Diastolic Blood Pressure (mmHg) Cardio 0 1	This violin plot shows that higher diastolic BP values are more common in individuals with cardiovascular disease, reinforcing that elevated blood pressure is a key heart disease risk factor.
 Boxplot of Age by Cardio Status Age (in days) Cardio 0 1	This boxplot graphs shows that Individuals with cardiovascular disease also tend to be older.
 Boxplot of Weight by Cardio Status Weight (in kg) Cardio 0 1	This boxplot graphs shows that Individuals with cardiovascular disease also tend to be heavier.

5.4 Models

5.4.1 Decision Tree Model

The dataset is split into 70% training and 30% testing dataset. To optimize the decision tree, hyperparameter tuning is performed using GridSearchCV. After hyperparameter optimisation, the best parameters for the model were identified as shown below:

Parameter	Value
Criterion	Entropy
Maximum tree depth	5
Minimum sample split	2
Number of samples at leaf node	1

5.4.2 Random Forest Model

For the random forest machine learning model, GridSearchCV was performed to set up hyperparameter tuning. Hyperparameter tuning was conducted using a 5-fold cross-validation approach. The optimal parameters for the model were determined to be as follow:

Parameter	Value
Number of trees	300
Maximum tree depth	10
Minimum sample split	10
Number of features	sqrt
Number of samples at leaf node	1

5.4.3 XGBoost Model

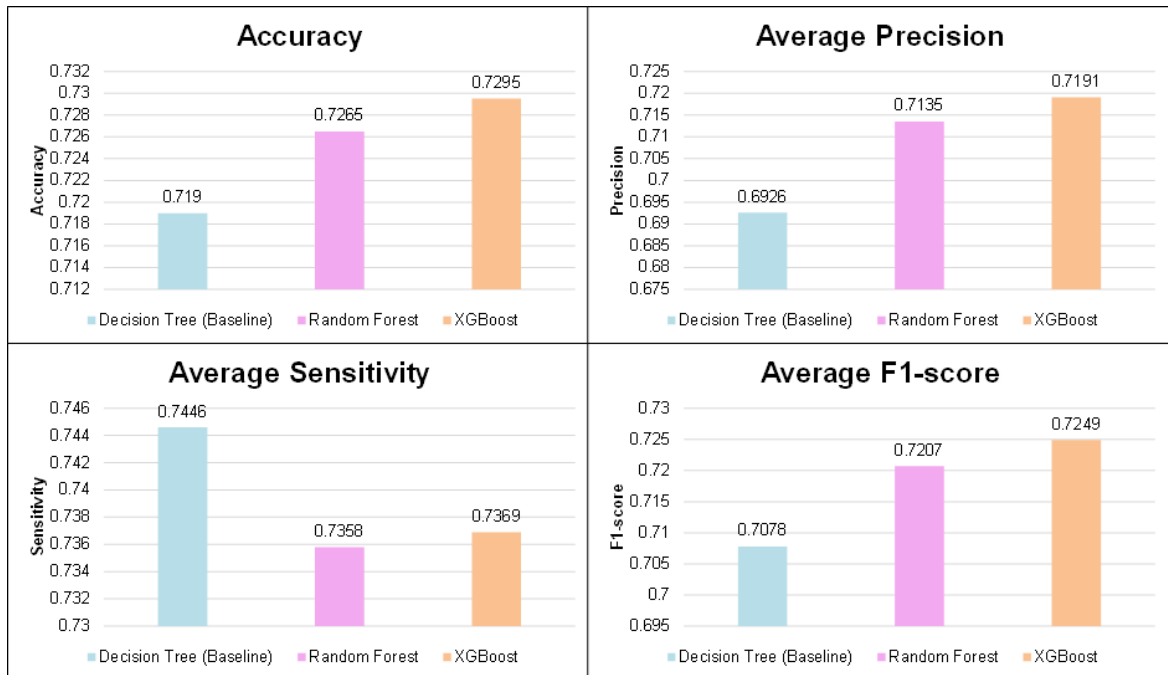
Firstly, hyperparameter tuning for the XGBoost model using GridSearchCV. A 5-fold cross-validation approach is employed during the grid search to evaluate the model's performance. After finding the best combination of hyperparameters, the model is retrained on the entire training dataset. The parameter grid includes:

Parameter	Value
Fraction of features	0.8
Gamma	0.2
Learning rate	0.01
Maximum tree depth	5

Number of estimators	300
L1 regularization	0
L2 regularization	1
Subsample	0.8

5.5 Results Interpretation

5.5.1 Comparison of the Results between Machine Learning Models



Machine Learning Model	True Positives	False Positives	False Negatives	True Negatives	Accuracy	Average Precision	Average Sensitivity	Average F1-Score
Decision Tree (Baseline Model)	8743	1718	4183	6356	0.719	0.6926	0.7446	0.7078
Random Forest	5334	1323	2269	4207	0.7265	0.7135	0.7358	0.7207
XGBoost	5300	1357	2195	4281	0.7295	0.7191	0.7369	0.7249

Based on the table and graph above, XGBoost has an **accuracy** of 72.95% which is closely followed by the random forest model with an accuracy of 72.62%%, and lastly the Decision Tree

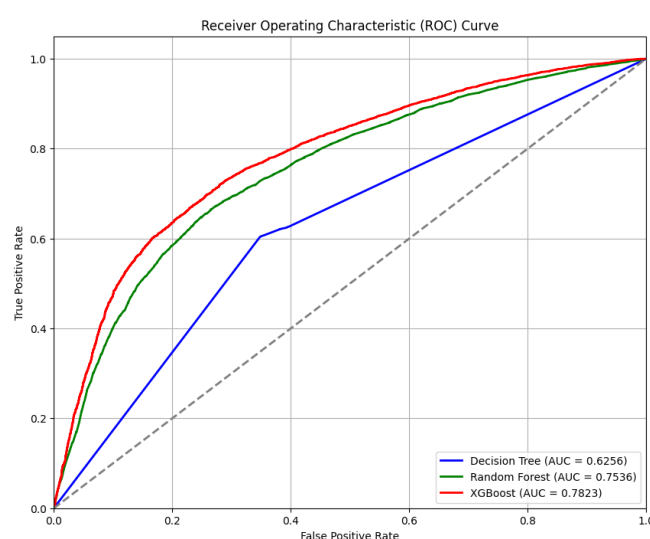
(Baseline) with an accuracy of 71.90%. This suggests that both Random Forest and XGBoost perform better than the baseline model due to the ensemble learning power of both methods.

XGBoost performs the best for overall **precision** with a score of 0.7191, slightly better than Random Forest at 0.7135 and much better than Decision Tree (Baseline) at 0.6926. This means that XGBoost is better at predicting positive predictions correctly, and minimizing false positives, and has better classifying power.

For the overall **sensitivity** shows that the Decision Tree (Baseline) performs best with a score of 0.7446 as it performs best in identifying true positives among both classes. Random Forest (0.7358) and XGBoost (0.7369) are slightly lower in this metric, implying some trade off in sensitivity in order to gain in precision, or other metrics.

XGBoost has the highest **F1-score** (0.7249), which shows that there is a good tradeoff between precision and recall. Random Forest comes very close with a score of 0.7207 and Decision Tree (Baseline) performs the lowest at 0.7078. It supports the overall superior performance of XGBoost: it balances false positives and false negatives effectively.

5.5.2 ROC-AUC Graph



A graphical representation of the trade-off between sensitivity (true positive rate) versus specificity (1 - false positive rate) for each model for varying classification thresholds is represented by the AUC-ROC curve. Among the three models, XGBoost achieved the highest AUC score (0.7823), as it had the highest performance of distinguishing positive (cardiovascular disease) and negative (no cardiovascular disease) classes.

A Decision Tree (Baseline) with the lowest AUC score (0.6256) was the lowest performer in the following and the lowest AUC score of the three models (Random Forest was 0.7536). Its ability to retain such high sensitivity with relatively few false positives is clearly demonstrated by the steeper initial slope of the XGBoost curve, which is the most effective model overall for this classification task. While Random Forest is robust, it marginally weakly interprets the ability to balance sensitivity and specificity as XGBoost. Being a simpler model, the Decision Tree always had a harder time with more pronounced trade-offs and thus had the lowest AUC score. Our

results emphasize the significance of the ensemble learning methods like XGBoost and Random Forest to attain a better classification performance.

Link to the Python notebook containing the machine learning models and results:

https://drive.google.com/drive/folders/1A-mMAh48hvNvqxXa1zLqgPyFMte4ZQFO?usp=drive_link

6.0 Plan for Reproducible Research

Reproducible research means a study that is said to be reproducible. Reproducible research is the practice of conducting and presenting research in such a way that others, and yourself can later re-implement your research strategy without ambiguity. In another words, Reproducibility refers to a different group of researchers arriving at the same results using their own data and methods. This section outlines a plan to ensure the reproducibility of our project. Data science is a collaborative endeavour as no data scientist knows all relevant modelling techniques and analyses. Hence, as a data scientist, we should be concerned on how we share and deploy our work for better collaboration. Therefore, others will be able to learn and help the whole team to improve together, making progress shared and beneficial for everyone, not just individuals.

6.1 Environment Setup and Programming Tools

In our project, Google Colab Notebook is used as our collaboration platform which all of our group member can work together and share the progress. Besides, we follow consistent and clear coding standards by using Python as our coding script. At the same time, we added comments and use meaningful variable names to increase code readability.

6.2 Version Control

For our version control, we use a simple method by saving the Google Colab Notebook to Google Drive or simple version tracking. Google Drive's "Version history" feature allows us to view and revert to earlier versions. Meanwhile, Colab notebooks automatically save and maintain version checkpoints during the work. Therefore, we can access the "File > Revision history" menu to see previous versions and revert if needed. We can also use comments in code cells to describe the updates and changes.

While Colab offers basic versioning and integration, it is not a full-fledged version control system. Therefore, for better version controlling, we will consider to put our work to Github as it enables robust version tracking using Git.

6.3 Online Repositories

For easy collaboration and complete view of the overall project, we will post our project on repository platforms like GitHub or Kaggle. Online repositories like Kaggle and GitHub are platforms that allow us to store, share, and collaborate on projects, especially data science, machine learning, and software development. In this way, we can get community feedback on our work and is able to improve our current work. Other than that, sharing work on GitHub or Kaggle benefits both the creator and the wider community, fostering a culture of collaboration and learning. By sharing techniques and approaches in solving problems, this may inspire others and produce new ideas.

6.4 Documentation

We ensure that our documents were labelled properly and can be easily understood to ensure reproducibility.

Project Overview

A clear and specific title was mentioned on our documents with concise description to ensure that other researchers or collaborators can easily understand the main idea of our project. Also, the project objectives and goals were also stated in the document so that others can know what we want to achieve through our project.

Data Details

This outlines where our data was obtained, how it was processed, and any assumptions or transformations applied.

Instructions and code explanation

In our document, we provide instructions for setting up, running, and testing the project. We also added comments in the code and wrote additional developer documentation to explain how each component works.

7.0 Deployment of Data Product

7.1 Data Product

This data product is a **web-based platform** developed to streamline the initial analysis and prediction of cardiac disease risk. Designed for use by insurance companies, the platform provides a user-friendly interface that is accessible on any device, anywhere. It allows insurance sales agents to input customer details directly into the platform and instantly generates a prediction of the customer's cardiac disease risk. This prediction enables insurers to determine whether a policy requires further review by a medical underwriter.

7.2 Workflow of Data Product

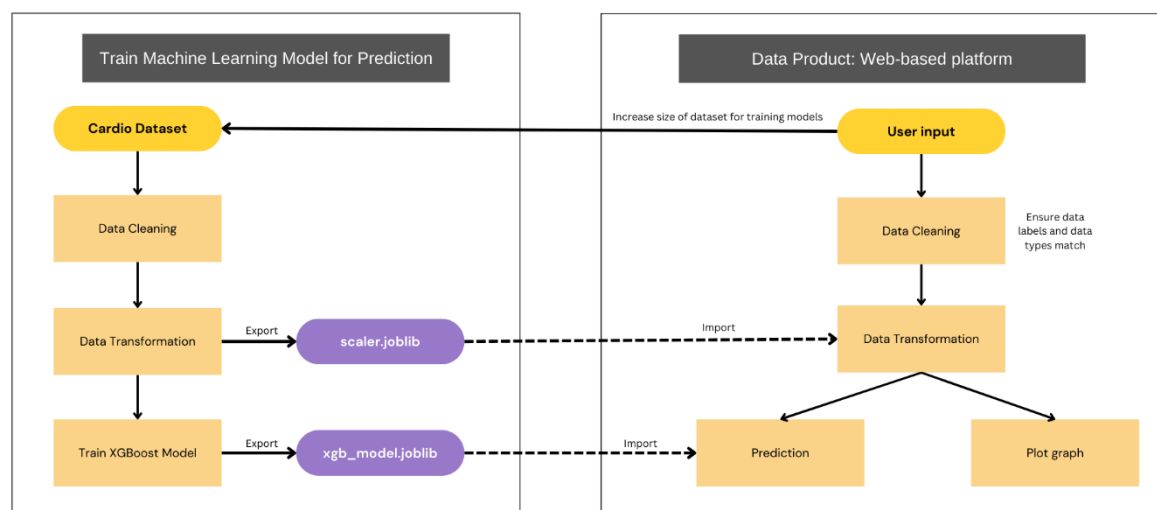


Figure 1. Workflow of Data Product

As shown in Figure 1, the workflow begins with training and testing the model using the `cardio_train.csv` dataset. During data-preprocessing, a Standard Z-score scaler is applied to transform the dataset, ensuring consistent scaling of numerical features. The fitted scaler is exported as `scaler.joblib` for later use in the data product. The best-performing model, XGBoost, is selected and exported as `xgb_model.joblib` making it ready for deployment.

In the deployed data product, when a user inputs data to generate predictions, the system first validates the datatype and labels of the inputs, adjusting them if necessary to ensure consistency with the model's requirements. The imported `scaler.joblib` scales the input data, which is then passed to `xgb_model.joblib` to generate predictions. The full dataset, combined with the input

data, is used to create visualizations such as a correlation plot between features and the target variable, as well as box plots comparing the input data to the dataset distribution. User inputs are stored in a separate dataset and periodically added to the original dataset to increase sample size, allowing for model retraining and updates.

7.3 User Interface (UI) Design

Cardiac Disease Prediction

Enter your health details to check the probability and risk level of cardiac disease.

Age (years)

50

Gender

☐ Female (1)
 ☒ Male (2)

Height (cm)

150

Weight (kg)

60

Systolic Blood Pressure (ap_hi)

120

Diastolic Blood Pressure (ap_lo)

90

Cholesterol

☐ Normal (1)
 ☒ Above Normal (2)
 ☐ Well Above Normal (3)

Glucose

☐ Normal (1)
 ☒ Above Normal (2)
 ☐ Well Above Normal (3)

Smoking

☐ No (0)
 ☒ Yes (1)

Alcohol Intake

☐ No (0)
 ☒ Yes (1)

Physical Activity

☐ Inactive (0)
 ☒ Active (1)

Predict

Probability of Cardiac Disease

0.2165

Risk Level

Low Risk

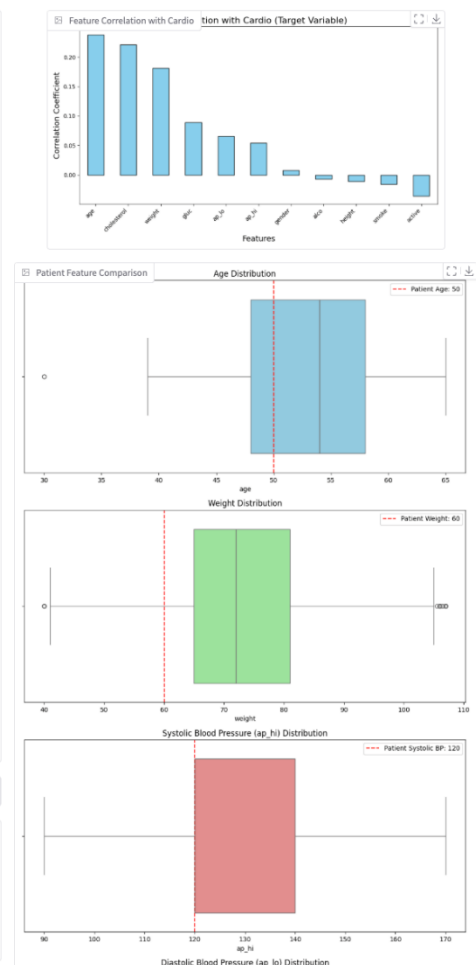


Figure 2. User Interface of the Data Product

As shown in Figure 2, the UI is divided into two main sections: **Input and Risk predictions (Left Side)** and **Visualization (Right Side)**.

1. Left Side: Input Form and Risk predictions

The top includes a structured form with labelled fields for age, body metrics, lifestyle habits, and other features. Radio buttons are used for categorical inputs (e.g., cholesterol levels, glucose levels) to eliminate invalid entries and prevent errors caused by inconsistent or incorrect data. At the bottom, the risk prediction section includes a "Predict" button that triggers the prediction process, resulting in the prediction of the probability of having cardiac disease, along with a simple "High Risk" or "Low Risk" output.

2. Right Side: Visualization and Results

The top section displays a **correlation plot** showing the relationships between input features (e.g., age, cholesterol, weight) and the target variable (cardiac disease). This allows users to observe which features contribute the most to the risk prediction, offering both transparency and interpretability.

The bottom section provides a **dynamic Patient Feature Comparison**, which is displayed after the "Predict" button is pressed. It shows box plots for selected features highly correlated with cardiac disease, visually comparing the customer's values with the dataset distribution. This comparison highlights where the customer stands relative to others.

8.0 Insights and Conclusion

Customized Insurance Plans: The insurance provider may provide high-risk applicants with specialized insurance plans that cater to their particular requirements, such as more extensive coverage for heart-related disorders or extra choices like preventative care coverage.

Targeted Health Interventions and Wellness Programs: The insurance company can plan and provide health intervention programs including quitting smoking, encouraging physical exercise, and improving diet for candidates who have been recognized as having a high risk of cardiovascular disease. Policyholders may be offered these programs as part of a wellness package or at a discounted fee.

Automated Approval Workflow: The risk assessment procedure may be automated by integrating the risk model into the insurance company web-based platform (data product). As a result, applications will be processed and decisions made more quickly. By flagging high-risk candidates for additional evaluation right away, manual assessments can take less time.

Conclusion

In this report, the data science process is performed according to the OSEMN framework. All the objective is achieved in this assignment and a web-based platform (data product) is created. Among three ML models, XGBoost model has the highest accuracy and precision of 72.95% and 0.7191. For the overall sensitivity, Decision Tree (Baseline) model performs best with a score of 0.7446. XGBoost has the highest F1-score of 0.7249. It supports the overall superior performance of XGBoost as it balances false positives and false negatives effectively. Based on the AUC-ROC curve, XGBoost achieved the highest AUC score of 0.7823, as it had the highest performance of distinguishing positive (cardiovascular disease) and negative (no cardiovascular disease) classes.

9.0 References

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